

This is a hint for **Exercise 6**.

Prop For any composition of linear maps $U \xrightarrow{f} V \xrightarrow{g} W$, one has the following induced mapping sequence

$$\left[\begin{array}{ccccccccccccccc} & x & \mapsto & x & & z & \mapsto & [z]_{\text{im } f} & & [u]_{\text{im } g \circ f} & \mapsto & [u]_{\text{im } g} & & \\ 0 & \xrightarrow{h_1} & \ker f & \xrightarrow{h_2} & \ker g \circ f & \xrightarrow{h_3} & \ker g & \xrightarrow{h_4} & \frac{V}{\text{im } f} & \xrightarrow{h_5} & \frac{W}{\text{im } g \circ f} & \xrightarrow{h_6} & \frac{W}{\text{im } g} & \xrightarrow{h_7} & 0 \end{array} \right],$$

satisfying $\ker h_i = \text{im } h_{i+1}$ for $1 \leq i \leq 6$.

In particular,

1. If g is injective, then $\ker f = \ker(g \circ f)$, and the following surjection has kernel $\frac{V}{\text{im } f}$:

$$\frac{W}{\text{im } g \circ f} \twoheadrightarrow \frac{W}{\text{im } g}, \quad (w + \text{im } g \circ f) \mapsto (w + \text{im } g);$$

2. If f is surjective, then $\frac{W}{\text{im}(g \circ f)} = \frac{W}{\text{im}(g)}$, and the following surjection has kernel $\ker f$:

$$\ker g \circ f \twoheadrightarrow \ker g, \quad v \mapsto f(v);$$

3. If $g \circ f$ is injective, then so is f ;

4. If $g \circ f$ is surjective, then so is g ;

5. The middle h_4 is factors through

$$\ker g \xrightarrow{\text{满}} \frac{\ker g}{\text{im } f \cap \ker g} \simeq \frac{\text{im } f + \ker g}{\text{im } f} \xrightarrow{\text{单}} \frac{U}{\text{im}(f)}.$$